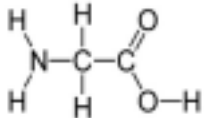


2(a)(i)	sequence of nucleotides or length / section / AW, of DNA ; A bases for nucleotides (a gene can) code for a polypeptide ; A protein for polypeptide	2
2(a)(ii)	 amino group added correctly ; A H ₂ N (remaining part of) carboxylic acid group added correctly ; A OH for O–H	2
2(a)(iii)	amylose <u>and</u> amylopectin ;	1

Question	Answer	Marks																				
2(b)	one mark for glucose and starch rows ; one mark for maltase row ; one mark for maltose row ; <table><tr><th>substance</th><th>polysaccharide</th><th>monosaccharide</th><th>macromolecule</th></tr><tr><td>glucose</td><td>x</td><td>✓</td><td>x</td></tr><tr><td>maltase</td><td>x</td><td>x</td><td>✓</td></tr><tr><td>maltose</td><td>x</td><td>x</td><td>x</td></tr><tr><td>starch</td><td>✓</td><td>x</td><td>✓</td></tr></table>	substance	polysaccharide	monosaccharide	macromolecule	glucose	x	✓	x	maltase	x	x	✓	maltose	x	x	x	starch	✓	x	✓	3
substance	polysaccharide	monosaccharide	macromolecule																			
glucose	x	✓	x																			
maltase	x	x	✓																			
maltose	x	x	x																			
starch	✓	x	✓																			
2(c)	any one valid suggestion ; e.g. easier to extract idea that microorganisms can be cultured in large quantities and produce large amounts of enzyme higher rate of reaction active over a greater temperature range	1																				
2(d)(i)	any three from: 1 increase in temperature increases kinetic energy of, molecules / particles / substrate / enzyme ; A moving around faster 2 (increase in temperature) increases rate of, successful collisions between enzyme and substrate / enzyme–substrate complex formation / substrate binding to active site ; OR at 48 °C (47 or 49) ref. to, optimum temperature / maximum enzyme activity / active sites binding substrate at maximum capacity / AW ; 3 (at temperatures higher than optimum) ref. to denaturation / loss of shape of active site (and decrease in activity) ; 4 AVP ; e.g. weaker / hydrogen, bonds break (and ref. to denaturation)	3																				

Question	Answer	Marks
2(d)(ii)	optimum / peak / max activity / at 100% <u>and</u> to right of original ; line beyond optimum must be to right and, reach vertical axis at 50% or below / reach horizontal axis ;	2

Question	Answer	Marks
----------	--------	-------